HERNANI SAMUEL DINIZ

Systems Programmer | Low-Level | Rust

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PROFILE

Systems programmer specializing in emulators, virtual machines, and language runtimes in Rust. Focused on modeling execution state with the type system to make invalid states unrepresentable. Interested in ISA design, bytecode interpreters, and bare-metal systems with memory safety.

TECHNICAL SKILLS

Primary language: Rust
Also familiar with: Python, C
Domains: emulation, ISA design, bytecode interpreters, bare-metal systems, memory safety
Tools: cargo, git, wasm-pack, no_std environments
Concepts: RISC-V RV32IMA ISA, privileged spec, ELF loading, WebAssembly

PROJECTS

RISC-V RV32IMA Emulator
Rust · Completed | GitHub | Live Demo (Linux in the browser)

- **Full RV32IMA ISA**: integer, multiply/divide, and atomic instructions; machine-mode privileged spec (traps, CSRs, MRET, WFI, ECALL).
- **Boots a real Linux 6.1.14** kernel (Buildroot no-mmu) and **FreeRTOS** bare-metal ELF images — Linux boot used as a correctness invariant at every development step.
- Emulated peripherals: UART 16550, CLINT with accurate mtime/mtimecmp and timer interrupts.
- Ported from a minimal C reference (mini-rv32ima) via incremental refactoring, with Linux boot validating each step — essentially open-heart surgery on a running system.
- Built-in analysis tools (own implementation): objdump-compatible disassembler, per-instruction execution tracer with register diffs, single-step mode.
- **Compiles to WebAssembly** — kernel embedded at compile time via `include_bytes!`, runs in any modern browser with no server required.

EDUCATION

B.Sc. in Software Engineering — Unicesumar
Feb 2021 – Feb 2025

ADDITIONAL INFORMATION

Languages: Portuguese (native), English — advanced technical reading (ISA specs, RFCs, official documentation)